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ORIGINAL ARTICLE

CATHETER-BASED VALVE OPERATIONS

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Selection for transcatheter versus surgical aortic valve replacement and mid-term survival: results of the AUTHEARTVISIT study

Johann Auer[a,†], Pavla Krotka[b,†], Berthold Reichardt[c], Denise Traxler[d,e,f], Ralph Wendt[g], Michael Mildner[h], Hendrik Jan Ankersmit[d,e,*,†] and Alexandra Graf[b,†]

[a] Department of Internal Medicine I with Cardiology and Intensive Care, St Josef Hospital Braunau, Braunau am Inn, Austria
[b] Center for Medical Data Science, Medical University of Vienna, Vienna, Austria
[c] Austrian Social Health Insurance Fund, Eisenstadt, Austria
[d] Clinic of Thoracic Surgery, Medical University of Vienna, Vienna, Austria
[e] Laboratory for Cardiac and Thoracic Diagnosis, Regeneration and Applied Immunology, Vienna, Austria
[f] Department of Oral and Maxillofacial Surgery, Medical University of Vienna, Vienna, Austria
[g] Department of Nephrology, Hospital St Georg Leipzig, Leipzig, Germany
[h] Department of Dermatology, Medical University of Vienna, Vienna, Austria

* Corresponding author. Department of Thoracic Surgery, Laboratory for Cardiac and Thoracic Diagnosis, Regeneration and Applied Immunology, Medical University of Vienna. Waehringer Gürtel 18-20, 1090 Vienna, Austria. Tel: +43-1-40400-67770; e-mail: Hendrik.ankersmit@meduniwien.ac.at (H.J. Ankersmit).

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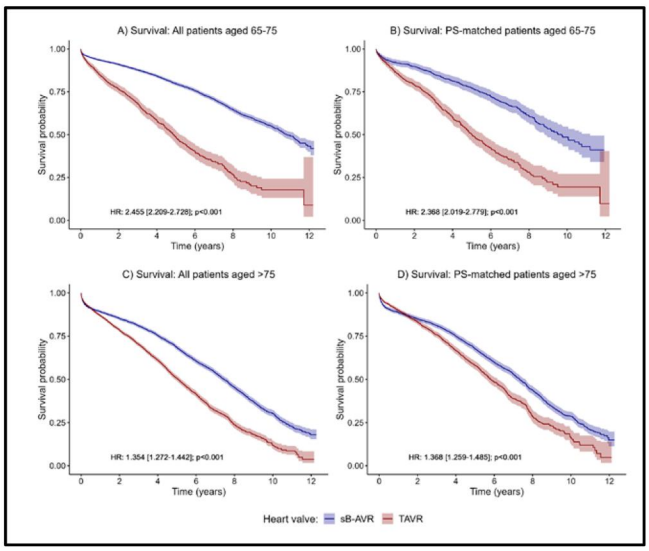
Selection for transcatheter versus surgical aortic valve replacement and mid-term survival: results of the AUTHEARTVISIT study

Summary

In patients ≥ 65 years aortic valve replacement with transcatheter aortic valve replacement (TAVR) was compared to surgically implanted bioprostheses (sB-AVR), both in the overall cohort and propensity score-matched populations (PSMP).Overall survival was significantly impaired in patients with TAVR between 65-75 years of age, both in the overall cohort (HR = 2.455, p < 0.001) and the PSMP cohort (HR = 2.386, p < 0.001).This finding was also present in patients > 75 years (overall: HR = 1.354, p < 0.001; PSMP: HR = 1.368, p < 0.001)



Legend: TAVR = transcatheter aortic valve replacement, sB-AVR = surgically implanted bioprostheses, PSMP = propensity score-matched populations, HR = hazard ratio

Abstract

OBJECTIVES: Limited data are available from randomized trials comparing outcomes between transcatheter aortic valve replacement (TAVR) and surgery in patients with different risks and with follow-up of at least 4 years or longer. In this large, population-based cohort

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study, long-term mortality and morbidity were investigated in patients undergoing aortic valve replacement (AVR) for severe aortic stenosis using a surgically implanted bioprosthesis (surgical/biological aortic valve replacement; sB-AVR) or TAVR.

METHODS: Individual data from the Austrian Insurance Funds from 2010 through 2020 were analysed. The primary outcome was all-cause mortality, assessed in the overall and propensity score-matched populations. Secondary outcomes included reoperation and cardiovascular events.

RESULTS: From January 2010 through December 2020, a total of 18 882 patients underwent sB-AVR ($n = 11\ 749$; 62.2%) or TAVR ($n = 7133$; 37.8%); median follow-up was 5.8 (95% CI 5.7–5.9) years (maximum 12.3 years). The risk of all-cause mortality was higher with TAVR compared with sB-AVR: hazard ratio 1.552, 95% confidence interval (CI) 1.469–1.640, $P < 0.001$; propensity score-matched hazard ratio 1.510, 1.403–1.625, $P < 0.001$. Estimated median survival was 8.8 years (95% CI 8.6–9.1) with sB-AVR versus 5 years (4.9–5.2) with TAVR. Estimated 5-year survival probability was 0.664 (0.664–0.686) with sB-AVR versus 0.409 (0.378–0.444) with TAVR overall, and 0.690 (0.674–0.707) and 0.560 (0.540–0.582), respectively, with propensity score matching. Separate subgroup analyses for patients aged 65–75 years and >75 years indicated a significant survival benefit in patients selected for sB-AVR in both groups. Other predictors of mortality were age, sex, previous heart failure, diabetes and chronic kidney disease.

CONCLUSIONS: In this retrospective national population-based study, selection for TAVR was significantly associated with higher all-cause mortality compared with sB-AVR in patients ≥65 years with severe, symptomatic aortic stenosis in the >2-year follow-up.

Keywords: Transcatheter aortic valve replacement • Bioprosthesis • AUTHEARTVISIT

ABBREVIATIONS

AVR	Aortic valve replacement
CI	Confidence interval
CMP	Cardiomyopathy
HR	Hazard ratio
ICD	International Classification of Diseases
MEL	Medizinische Einzelleistung
PSM	Propensity score matching
RCT	Randomized controlled trial
SAVR	Surgical aortic valve replacement
sB-AVR	Surgical/biological aortic valve replacement
TAVR	Transcatheter aortic valve replacement

INTRODUCTION

The comparative effectiveness of aortic valve replacement (AVR) for severe aortic stenosis using a surgically implanted bioprosthesis (surgical/biological aortic valve replacement; sB-AVR) or transcatheter aortic valve replacement (TAVR) can vary depending on several factors [1]. In general, surgical AVR (SAVR) is considered to be the gold standard for AVR because it provides a longer-lasting and more durable solution compared with TAVR. However, SAVR requires open-heart surgery, which is associated with a longer recovery time and higher risks of complications compared with minimally invasive TAVR. Both SAVR and TAVR have advantages and disadvantages, and factors involved in procedure selection include patient characteristics, type of valve and technical aspects of the procedure. TAVR and SAVR differ in length of hospital stay and overall recovery time, paravalvular regurgitation, haemodynamics, ease of coronary access, structural valve deterioration and need for a new pacemaker. Rates of vascular complications, pacemaker implantations and paravalvular regurgitation are higher after TAVR. In contrast, severe bleeding events, acute kidney injury and new-onset atrial fibrillation are more frequent after SAVR [2].

TAVR has emerged as the most frequently used therapy for older patients with severe symptomatic aortic stenosis and has surpassed SAVR in procedural volume in patients aged 75 years or older in the USA and Europe [3]. Randomized controlled trials (RCTs) have demonstrated equivalent or even superior outcomes with TAVR compared with SAVR in elderly patients across a variety of risk categories [4].

RCTs are usually the accepted gold standard for establishing the efficacy of medical interventions [1], but they cannot capture every aspect of a therapeutic intervention's effects in clinical practice [2, 3]. A systematic review and meta-analysis, for example, showed that RCTs comparing TAVR with SAVR had systematic imbalances in the proportion of deviations from random assigned treatment, loss to follow-up and receipt of additional procedures, including coronary artery bypass surgery. These factors can undermine internal validity because of the related high risk of attrition and performance biases [5]. To overcome such shortcomings of RCTs, real-world evidence is derived from studies conducted with non-randomized data, including data collected by the healthcare system, such as longitudinal insurance claims [6]. Such evidence can complement clinical trial findings, facilitating a more complete understanding of the effectiveness of therapeutic interventions.

The current analysis is based on data from the AUTHEARTVISIT study including patients with severe aortic stenosis who had an indication for AVR and were selected to undergo either TAVR or sB-AVR. The results characterize the mid-term and long-term clinical outcomes for this patient group, followed for longer than 4 years.

METHODS

Study design and patients

This retrospective national population-based study complied with the Declaration of Helsinki and was approved by the ethics committee of lower Austria (GS1-EK-4/722-2021). The trial was registered with ClinicalTrials.gov (NCT05912660). Study data were generated retrospectively by retrieval from the Austrian Health Insurance Funds. Data on outcomes and potential confounding factors were derived from billing information based on MEL (i.e. Medizinische Einzelleistung, or individual medical procedure) and International Classification of Diseases (ICD) codes available for each patient from 1 year before surgery up to study cut-off. Healthcare in Austria is a national system with good access to care, few malpractice lawsuits and little tendency towards

overuse of medical resources. It is based on a social insurance model founded on compulsory insurance. Access to individual services is regulated by social insurance law. All of the insured have a legal right to services, which are financed on the basis of solidarity. Austrian social insurance is based on the principle of solidarity and self-administration and is mainly financed via social insurance contributions. Approximately 98% of the Austrian population is registered in the public health insurance system, but a minority of patients paying for medical supplies using private insurance [7] were not included in the AUTHEARTVISIT trial.

This analysis included all patients aged ≥65 years registered in the Austrian Health System who underwent AVR with either sB-AVR or TAVR and without coronary revascularization within 4 months before the index procedure in Austria from 1 January 2010 through 31 December 2020. Patients who underwent AVR with a mechanical valve, had concomitant heart surgery or underwent percutaneous coronary stenting within 4 months prior to the index operation were excluded.

Study outcomes

The primary study outcome was all-cause mortality. Secondary outcomes were reoperation, stroke, myocardial infarction and heart failure. For each patient, billing information (based on MEL codes) and diagnoses (based on ICD codes) were available from 1 year before the index procedure up to study cut-off. Furthermore, death dates were available at the time of study cut-off. To evaluate secondary outcomes for each patient, data were scanned from the index procedure through the study cut-off date for the corresponding ICD or MEL codes. A detailed list of all ICD codes used for the outcome variables is shown in the supplementary file (Supplementary Material, Tables S1 and S2).

Statistical analysis

Continuous data are presented as means ± standard deviation or as medians with interquartile ranges. Categorical data are presented as absolute numbers and percentages. Continuous variables were compared between study groups using Student's *t*-test, and categorical variables were compared using the chi-squared test. The median follow-up-time was estimated using the reverse Kaplan–Meier method.

To evaluate the association between intervention (sB-AVR or TAVR) and the primary outcome of all-cause mortality, Cox regression was used. To account for potential confounding variables, multivariable Cox regression analysis was performed, with the following variables included: age (per 1 year increase), sex (male, female), heart failure, myocardial infarction, stroke, diabetes mellitus, adiposity hyperlipidaemia, hyperuricaemia/gout, valvular/arrhythmogenic/other cardiomyopathies (CMPs), ischaemic CMP, atherosclerosis, pulmonary diseases, kidney diseases or malignant diseases prior to the index operation. These comorbidities were defined using ICD codes available for each patient up to 1 year before the index procedure. Data available before the index procedure were scanned for each patient based on ICD codes and categorized for the different comorbidities. If an ICD code for a comorbidity as a main or secondary diagnosis was observed at least 1 time in the year before the index procedure, the patient was assumed to have the comorbidity. Detailed information on ICD codes used for this analysis is shown in the supplementary file (Supplementary Material, Tables S3–S5). Schönfeld residuals were used to evaluate the proportional hazard assumption and variance inflation factors to evaluate multicollinearity. Hazard ratios (HRs) are presented as TAVR versus sB-AVR, so that a HR >1 indicates increased risk of the corresponding event in the TAVR group (and vice versa).

Furthermore, propensity score matching (PSM) was performed for the 2 study groups (sB-AVR and TAVR). Propensity scores were estimated with logistic regression based on age, sex, heart failure, myocardial infarction, stroke, diabetes mellitus, adiposity, hyperlipidaemia, hyperuricaemia/gout, valvular/arrhythmogenic/other CMPs, ischaemic CMP, atherosclerosis, pulmonary diseases, kidney diseases and malignant diseases prior to the operation. Matching was performed using the nearest neighbour method (1:1 ratio with a calliper width of 0.0001 or 0.001 depending on the heterogeneity of the investigated subgroup).

To investigate results on all-cause mortality in more detail, several sensitivity analyses were performed. Given indications of a non-proportional hazard behaviour for the group variable (crossing survival curves in the PSM groups), we additionally investigated a potential differential effect of group on all-cause death in different time intervals (years 1, 2, 3, 4 and >4) using time-dependent coefficients for the study group. Furthermore, the primary Cox regression analyses for all-cause mortality were repeated for several subgroups of patients: ages 65–75 years, age >75 years, with diabetes mellitus, with pulmonary diseases, with kidney disease, with the index procedure before 2015, and with index procedure during and after 2015. Furthermore, patients surviving the first month after the index procedure were assessed for risk of all-cause mortality based on implantation or not of a pacemaker within that first month.

For the secondary outcomes (reoperation, myocardial infarction, heart failure, stroke), competing risk analyses were performed, using death as a competing risk. Confounding variables for multivariable analysis were the same as those input into the Cox regression model for the primary outcome. In case of heart failure, the analysis was adapted, as patients with a prior diagnosis of heart failure cannot be newly diagnosed and were therefore excluded.

All tests were performed using a two-sided significance level of 0.05. Because of the retrospective and exploratory character of the study, there was no correction for multiple comparisons. Statistical analyses were conducted and graphs generated in R (version 4.1.3) using the following packages: ggplot2 (version 3.4.2), survival (version 3.2.13), survminer (version 0.4.9), cmprsk (version 2.2.11) and MatchIt (version 4.4.0).

RESULTS

Study population and patient characteristics

The study included 18 882 patients registered in the Austrian Health Insurance Funds who underwent AVR with sB-AVR or TAVR from 1 January 2010 through 31 December 2020. sB-AVR was performed in 11 749 (62.2%) and TAVR in 7133 (37.8%). In 2010, sB-AVR was performed in most patients, but over time, TAVR became the more common choice, and in 2019, more patients received TAVR than sB-AVR (Fig. 1). We also divided the cohort into patients aged 65–75 years ($n = 7575$, 40.1%) and those aged >75 years ($n = 11\ 307$, 59.9%). In the younger group, sB-AVR was predominant ($n = 6483$, 85.6% versus $n = 1092$, 14.4% receiving TAVR), but not in the older group ($n = 5266$, 46.6% receiving

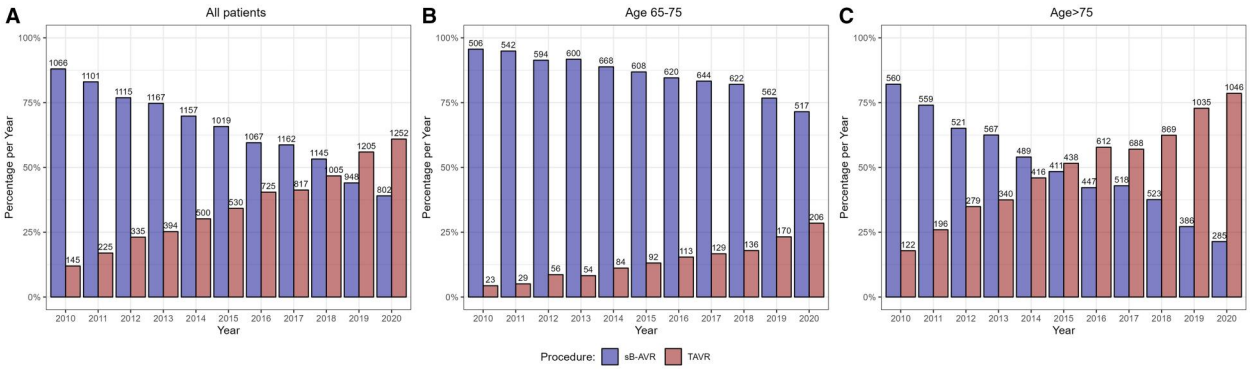

Figure 1: Percentages per year (*y*-axis) and absolute numbers (above the bars) of patients receiving TAVI or sB-AVR per calendar year for (**A**) the overall cohort, (**B**) those ages 65–75 years and (**C**) those older than 75 years. sB-AVR: surgical/biological aortic valve replacement.

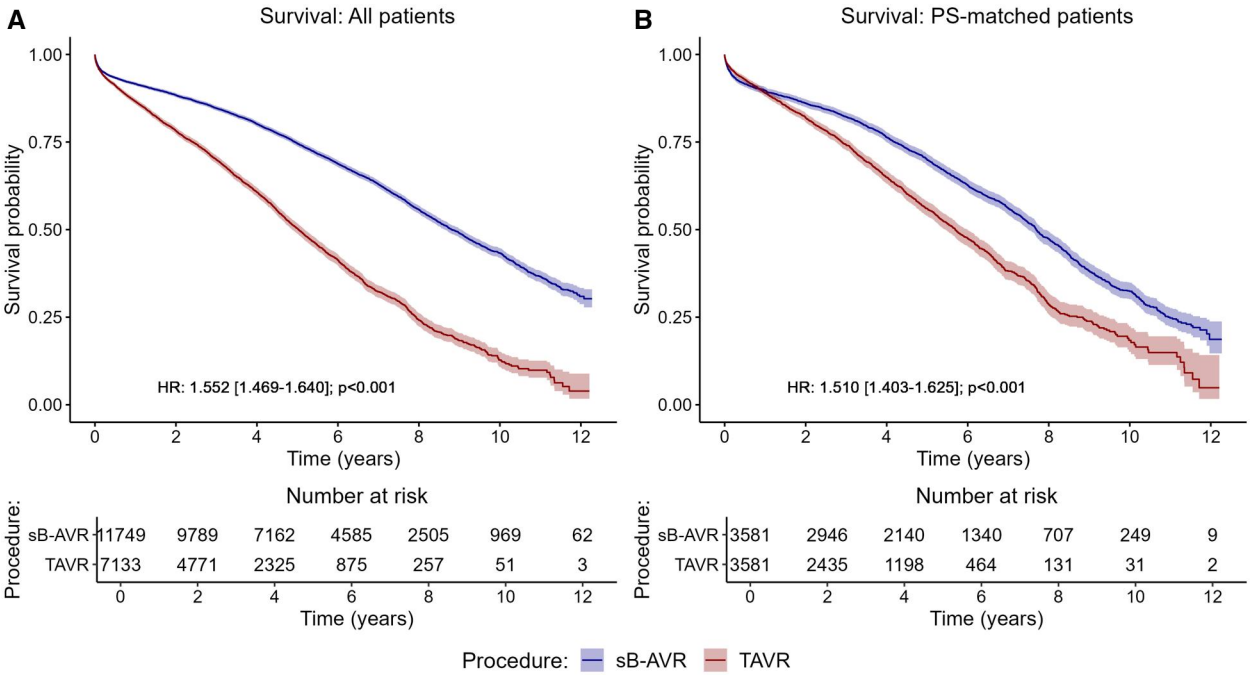

Figure 2: Kaplan–Meier curves and 95% confidence intervals for overall survival in (**A**) all patients and (**B**) the propensity score-matched groups. sB-AVR: surgical/biological aortic valve replacement; TAVR: transcatheter aortic valve replacement.

sB-AVR versus $n = 6041$, 53.4%, receiving TAVR). In the younger patient group, sB-AVR was the more common method throughout the follow-up period, in contrast to the older group, in which TAVR became more common in 2015 (Fig. 1).

All-cause mortality

The primary end-point was all-cause mortality. Kaplan–Meier curves for all patients (Fig. 2A) and PSM patients (Fig. 2B) showed a significantly increased all-cause mortality with TAVR compared to sB-AVR. Table 1 shows baseline characteristics between both groups before and after PSM. Also standard Cox regression, accounting for several confounding factors, showed an increased all-cause mortality with TAVR [HR 1.552, 95% confidence interval (CI) 1.469–1.640, $P < 0.001$] overall and in the PSM analysis (HR 1.510, 1.403–1.625, $P < 0.001$).

The choice of the valve replacement procedure was only 1 factor associated with survival, and also for other possible confounders significant results for all-cause mortality in the overall and PSM analyses were observed (Table 2, standard Cox model). Median follow-up was 5.8 (95% CI 5.7–5.9) years in the overall population, 4.2 (95% CI 4.1–4.6) years with TAVR and 6.8 (95% CI 6.7–6.9) years in the sB-AVR group. Estimated median survival time was 8.8 years (95% CI 8.6–9.1) in the sB-AVR group, compared with 5 years (4.9–5.2) in the TAVR group. In more detail, the estimated 5-year survival probability was 0.664 (95% CI 0.664–0.686) in the sB-AVR group, compared with 0.409 (0.378–0.444) in the TAVR group. PSM changed this estimate to 0.690 (0.674–0.707) with sB-AVR versus 0.560 (0.540–0.582) with TAVR.

The results suggested a varying effect of study group on the risk of all-cause mortality over time (crossing survival curves in the PSM patients in the 1st year after the index procedure; Fig. 2B). For this reason, we accounted for a potential time-varying trend using a Cox model with time-dependent coefficients for the grouping factor (Table 2, time-dependent analyses). The analyses of all patients and of PSM patients in the

Table 1: Baseline characteristics compared between the sB-AVR and TAVR groups before and after propensity score matching (PSM)

	All data				After PSM			
	sB-AVR ($n=11749$)		TAVR ($n=7133$)		sB-AVR ($n=3581$)		TAVR ($n=3581$)	
	n	Estimate (CL)	*n*	Estimate (CL)	*n*	Estimate (CL)	*n*	Estimate (CL)
Age (years), mean (CI)	11749	74.7 (74.6–74.8)	7133	81.1 (81.0–81.2)	3581	78.7 (78.6–78.9)	3581	78.7 (78.5–78.8)
Sex								
Male	6886	58.61 (57.71–59.5)	3177	44.54 (43.38–42.29)	1738	48.53 (46.89–50.19)	1742	48.65 (47–53.11)
Female	4863	41.39 (40.50–42.29)	3956	55.46 (54.30–56.62)	1843	51.47 (49.81–53.11)	1839	51.35 (49.7–53)
Heart failure[a]								
No	10215	86.94 (86.32–87.54)	5437	76.22 (75.21–13.68)	3072	85.79 (84.59–86.9)	3039	84.86 (83.64–15.41)
Yes	1534	13.06 (12.46–13.68)	1696	23.78 (22.80–24.79)	509	14.21 (13.1–15.41)	542	15.14 (13.99–16.36)
Myocardial Infarction[a]								
No	11332	96.45 (96.10–96.77)	6989	97.98 (97.62–3.90)	3540	98.86 (98.43–99.17)	3540	98.86 (98.43–1.57)
Yes	417	3.55 (3.23–3.90)	144	2.02 (1.71–2.38)	41	1.14 (0.83–1.57)	41	1.14 (0.83–1.57)
Stroke[a]								
No	11557	98.37 (98.12–98.58)	6997	98.09 (97.74–1.88)	3561	99.44 (99.12–99.65)	3563	99.5 (99.19–0.88)
Yes	192	1.63 (1.42–1.88)	136	1.91 (1.61–2.26)	20	0.56 (0.35–0.88)	18	0.5 (0.31–0.81)
Diabetes mellitus[a]								
No	9632	81.98 (81.27–82.67)	5718	80.16 (79.21–18.73)	3047	85.09 (83.87–86.23)	3025	84.47 (83.24–16.13)
Yes	2117	18.02 (17.33–18.73)	1415	19.84 (18.92–20.79)	534	14.91 (13.77–16.13)	556	15.53 (14.36–16.76)
Adiposity[a]								
No	10965	93.33 (92.86–93.77)	6718	94.18 (93.61–7.14)	3458	96.57 (95.9–97.13)	3431	95.81 (95.09–4.10)
Yes	784	6.67 (6.23–7.14)	415	5.82 (5.29–6.39)	123	3.43 (2.87–4.10)	150	4.19 (3.57–4.91)
Hyperlipidaemia[a]								
No	8549	72.76 (71.95–73.57)	5366	75.23 (74.21–28.05)	2793	77.99 (76.59–79.34)	2775	77.49 (76.08–23.41)
Yes	3200	27.24 (26.43–28.05)	1767	24.77 (23.78–25.79)	788	22.01 (20.66–23.41)	806	22.51 (21.16–23.92)
Hyperuricaemia/gout[a]								
No	11325	96.39 (96.03–96.72)	6851	96.05 (95.56–3.97)	3528	98.52 (98.05–98.88)	3537	98.77 (98.34–1.95)
Yes	424	3.61 (3.28–3.97)	282	3.95 (3.52–4.44)	53	1.48 (1.12–1.95)	44	1.23 (0.90–1.66)
Valvular, rhythmological, other cardiomyopathies[a]								
No	1694	14.42 (13.79–15.07)	606	8.50 (7.86–86.21)	412	11.51 (10.49–12.61)	418	11.67 (10.65–89.51)
Yes	10055	85.58 (84.93–86.21)	6527	91.50 (90.83–92.14)	3169	88.49 (87.39–89.51)	3163	88.33 (87.22–89.35)
Ischaemic cardiomyopathy[a]								
No	5385	45.83 (44.93–46.74)	3379	47.37 (46.21–55.07)	1690	47.19 (45.55–48.84)	1679	46.89 (45.24–54.45)
Yes	6364	54.17 (53.26–55.07)	3754	52.63 (51.46–53.79)	1891	52.81 (51.16–54.45)	1902	53.11 (51.46–54.76)
Atherosclerosis[a]								
No	11055	94.09 (93.65–94.51)	6531	91.56 (90.89–6.35)	3432	95.84 (95.12–96.46)	3428	95.73 (95–4.88)
Yes	694	5.91 (5.49–6.35)	602	8.44 (7.81–9.11)	149	4.16 (3.54–4.88)	153	4.27 (3.64–5.00)
Pulmonary diseases[a]								
No	11398	97.01 (96.68–97.31)	6678	93.62 (93.02–3.32)	3510	98.02 (97.49–98.44)	3502	97.79 (97.24–2.51)
Yes	351	2.99 (2.69–3.32)	455	6.38 (5.83–6.98)	71	1.98 (1.56–2.51)	79	2.21 (1.76–2.76)
Kidney diseases[a]								
No	10546	89.76 (89.20–90.30)	5655	79.28 (78.32–10.8)	3203	89.44 (88.38–90.42)	3189	89.05 (87.97–11.62)
Yes	1203	10.24 (9.70–10.80)	1478	20.72 (19.79–21.68)	378	10.56 (9.58–11.62)	392	10.95 (9.95–12.03)
Malignant diseases[a]								
No	11194	95.28 (94.87–95.65)	6682	93.68 (93.08–5.13)	3469	96.87 (96.23–97.41)	3449	96.31 (95.63–3.77)
Yes	555	4.72 (4.35–5.13)	451	6.32 (5.77–6.92)	112	3.13 (2.59–3.77)	132	3.69 (3.10–4.37)

[a]Present before operation; Estimates are % (95% confidence interval) unless otherwise specified.

CI: Confidence intervals; sB-AVR: surgical/biological aortic valve replacement; TAVR: transcatheter aortic valve replacement.

1st year after the index procedure showed no significant difference between TAVR and sB-AVR (all patients: HR 1.073, 95% CI 0.977–1.178, $P=0.14$; PSM patients: 1.044, 0.905–1.203, $P=0.56$; Table 2). However, for follow-up times >1 year, as already found in the standard Cox model, a significantly higher risk for all-cause mortality was found in patients receiving TAVR, indicating that improved survival with sB-AVR may be evident only with longer follow-up times.

The effect of a significantly higher risk for all-cause mortality in TAVR as compared with sB-AVR was even more prominent in patients aged 65–75 years (Fig. 3A and B; HR 2.455, 95% CI 2.209–2.728, $P<0.001$; Fig. 3A). This effect was also observed in the PSM group (HR 2.368, 2.019–2.779, $P<0.001$; Fig. 3B). In the time-dependent analyses, across all separately investigated time intervals, a significantly better survival was found (Supplementary Material, Table S6). Details on baseline data for the subgroup of patients aged 65–75 years and the corresponding PSM patients are given in Supplementary Material, Table S7.

Of note, in older patients (>75 years), TAVR performed significantly worse for the primary outcome of all-cause mortality overall (HR 1.345, 95% CI 1.272–1.442, $P<0.001$; Fig. 3C) and in the PSM patients (1.368, 1.259–1.485, $P<0.001$; Fig. 3D). The time-dependent analyses again showed no significant difference between groups in the 1st year after the index procedure but a significantly better survival in the sB-AVR group with longer follow-up times (Supplementary Material, Table S8). Details on

Table 2: Hazard ratios (HRs) and corresponding 95% confidence intervals (CIs) from multivariable Cox regression accounting for all listed confounder for all-cause mortality in all patients and the propensity score-matched patients for the standard and time-dependent models (see statistical methods section)

	Standard Cox model				Time-dependent analyses			
	All patients		Propensity score–matched		All patients		Propensity score–matched	
	HR (95% CI)	*P*-value	HR (95% CI)	*P*-value	HR (95% CI)	*P*-value	HR (95% CI)	*P*-value
AVR procedure (sB-AVR versus TAVR)	1.552 (1.469–1.640)	**<0.001**	1.510 (1.403–1.625)	**<0.001**				
Year 1					1.073 (0.977–1.178)	**0.14**	1.044 (0.905–1.203)	**0.56**
Year 2					1.902 (1.662–2.176)	**<0.001**	2.006 (1.615–2.492)	**<0.001**
Year 3					1.746 (1.520–2.005)	**<0.001**	1.971 (1.589–2.445)	**<0.001**
Year 4					1.796 (1.563–2.064)	**<0.001**	1.704 (1.393–2.083)	**<0.001**
Year >5					1.750 (1.611–1.901)	**<0.001**	1.574 (1.397–1.774)	**<0.001**
Age (per 1 year increase)	1.053 (1.049–1.058)	**<0.001**	1.036 (1.028–1.044)	**<0.001**	1.053 (1.049–1.058)	**<0.001**	1.036 (1.028–1.044)	**<0.001**
Sex (male versus female)	0.865 (0.826–0.906)	**<0.001**	0.872 (0.812–0.937)	**<0.001**	0.863 (0.824–0.904)	**<0.001**	0.872 (0.812–0.937)	**<0.001**
Heart failure[a]	1.420 (1.343–1.502)	**<0.001**	1.373 (1.248–1.511)	**<0.001**	1.424 (1.346–1.506)	**<0.001**	1.375 (1.250–1.513)	**<0.001**
Myocardial Infarction[a]	1.166 (1.032–1.317)	**0.01**	1.427 (1.075–1.894)	**0.01**	1.158 (1.025–1.308)	**0.02**	1.425 (1.074–1.891)	**0.01**
Stroke[a]	1.214 (1.031–1.428)	**0.02**	0.952 (0.573–1.583)	**0.85**	1.210 (1.029–1.424)	**0.02**	0.951 (0.572–1.581)	**0.85**
Diabetes mellitus[a]	1.379 (1.303–1.460)	**<0.001**	1.299 (1.177–1.434)	**<0.001**	1.381 (1.305–1.461)	**<0.001**	1.299 (1.177–1.434)	**<0.001**
Adiposity[a]	1.118 (1.021–1.225)	**0.02**	1.118 (0.936–1.336)	**0.22**	1.116 (1.019–1.222)	**0.02**	1.118 (0.936–1.335)	**0.22**
Hyperlipidemia[a]	0.825 (0.781–0.871)	**<0.001**	0.822 (0.750–0.900)	**<0.001**	0.824 (0.781–0.871)	**<0.001**	0.822 (0.750–0.900)	**<0.001**
Hyperuricaemia/gout[a]	1.089 (0.972–1.219)	0.14	1.121 (0.841–1.496)	**0.44**	1.091 (0.974–1.222)	**0.13**	1.120 (0.839–1.493)	**0.44**
Valvular, rhythmological, other cardiomyopathies[a]	0.865 (0.804–0.931)	**<0.001**	0.837 (0.748–0.937)	**0.002**	0.866 (0.804–0.932)	**<0.001**	0.838 (0.748–0.938)	**0.002**
Ischaemic cardiomyopathy[a]	1.079 (1.029–1.132)	**0.002**	1.122 (1.040–1.209)	**0.003**	1.079 (1.029–1.132)	**0.002**	1.121 (1.040–1.208)	**0.003**
Atherosclerosis[a]	1.205 (1.109–1.308)	**<0.001**	1.229 (1.043–1.448)	**0.01**	1.204 (1.109–1.307)	**<0.001**	1.227 (1.042–1.446)	**0.01**
Pulmonary diseases[a]	1.310 (1.167–1.470)	**<0.001**	1.263 (0.970–1.645)	**0.08**	1.330 (1.186–1.493)	**<0.001**	1.267 (0.972–1.650)	**0.08**
Kidney diseases[a]	1.406 (1.322–1.496)	**<0.001**	1.508 (1.352–1.681)	**<0.001**	1.410 (1.325–1.500)	**<0.001**	1.508 (1.353–1.682)	**<0.001**
Malignant diseases[a]	1.424 (1.306–1.553)	**<0.001**	1.354 (1.136–1.614)	**<0.001**	1.428 (1.310–1.557)	**<0.001**	1.359 (1.140–1.620)	**<0.001**

[a]Present before operation.

sB-AVR: surgical/biological aortic valve replacement; TAVR: transcatheter aortic valve replacement.

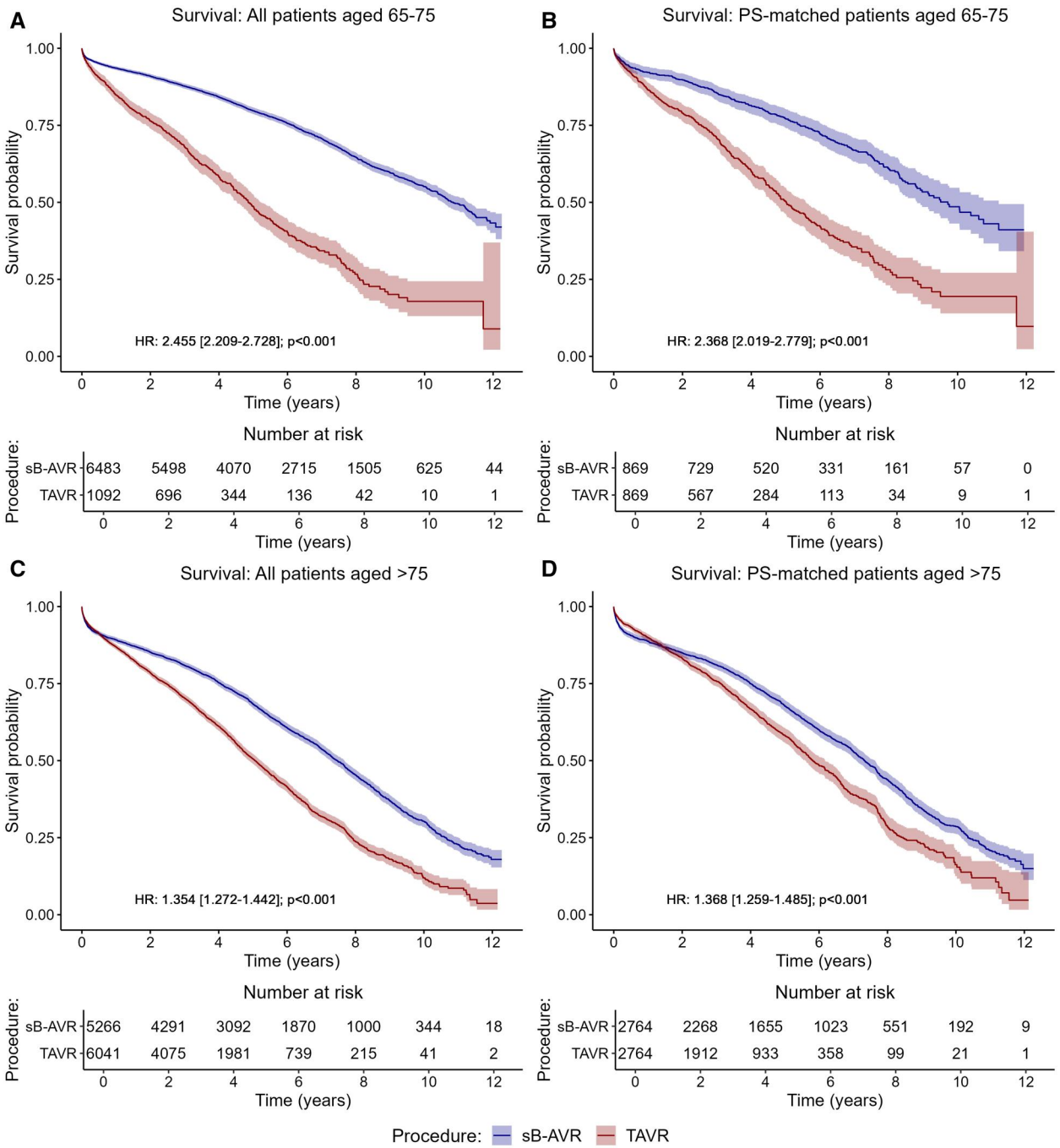

Figure 3: Kaplan–Meier curves and 95% confidence intervals for overall survival for subgroups of patients aged 65–75 years in the (**A**) overall cohort and (**B**) PSM groups and those aged >75 years in the (**C**) overall cohort and (**D**) PSM groups. sB-AVR: surgical/biological aortic valve replacement; PSM: propensitiy score matching; TAVR: transcatheter aortic valve replacement.

baseline data for the subgroup of patients >75 years and corresponding PSM patients can be found in Supplementary Material, Table S9.

All multivariable models observed that the choice of AVR is only 1 factor in a complex system (Table 2, Supplementary Material, Tables S6 and S8). To evaluate the results in more detail, we performed several sensitivity analyses (for detailed results, see Supplementary Material, file, chapters 4–6). In separate analyses of all-cause mortality in patients treated before 2015 as well as during and after 2015, TAVR patients still had a significant larger risk for all-cause mortality as compared to sB-AVR in both subsets (before 2015: HR 1.570, 95% CI 1.456–1.690, $P < 0.001$; during/after 2015: HR 1.704, 1.562–1.860, $P < 0.001$; Supplementary Material, Fig. S1, Supplementary Material, Table S10).

Furthermore, we investigated all-cause mortality by intervention type in 3 patient subsets defined by underlying disease (diabetes, pulmonary diseases, kidney diseases). In all 3 subsets, all-cause mortality was significantly increased in patients with TAVR as the primary intervention. Of the 3532 patients with diabetes before the intervention, 1735 died within the follow-up period (sB-AVR, $n = 988$; TAVR, $n = 747$). TAVR as the primary intervention was

associated with significantly higher risk for all-cause mortality (HR 1.587, 1.418–1.777, $P < 0.001$; Supplementary Material, Fig. S2A; Supplementary Material, Table S11). The subset with pulmonary diseases was notably smaller at 806 patients, of whom 108 in the sB-AVR group and 204 in the TAVR group died. For this patient group, TAVR was significantly associated with survival larger risk for all-cause mortality (HR 2.057, 1.556–2.720, $P < 0.001$; Supplementary Material, Fig. S2B, Supplementary Material, Table S11). In the 2681 patients with kidney diseases at the date of the primary intervention, 606 in the sB-AVR group and 813 in the TAVR group died during follow-up (TAVR versus sB-AVR, HR 1.436, 1.269–1.625, $P < 0.001$; Supplementary Material, Fig. S2C, Supplementary Material, Table S11).

Pacemaker implantation is common after AVR, so we also analysed all-cause mortality in patients with implantation within the 1st month after AVR (Supplementary Material, Fig. S3, Supplementary Material, Table S12). A total of 662 patients were excluded from this subanalysis (394 sB-AVR and 268 TAVR) because they died within the 1st month after the procedure and could not have received a pacemaker. Patients who underwent TAVR had a higher incidence of permanent pacemaker implantation (11.22%) compared with those who had sB-AVR (4.5%). Patients receiving a pacemaker had a significantly higher risk of all-cause death compared with patients who did not receive a pacemaker (HR 1.107, 1.016–1.207, $P = 0.02$).

Secondary outcomes

Of the 287 patients needing aortic valve reintervention, 232 had undergone sB-AVR and 55 TAVR. Risk of reoperation was not significantly increased by intervention type (HR 0.822, 95% CI 0.597–1.132, $P = 0.23$) in the overall patient cohort (Supplementary Material, Fig. 4A, Supplementary Material, Table S13).

As noted, not all patients were included in the analyses for newly diagnosed heart failure. Of the 15 652 patients who were included, 3671 in the overall cohort developed heart failure, 2339 had sB-AVR and 1332 had TAVR. The risk of developing heart failure did not differ significantly with valve type (HR 1.030, 0.949–1.118, $P = 0.47$; Supplementary Material, Fig. 4B, Supplementary Material, Table S13).

During the study follow-up, 443 patients had a myocardial infarction, 302 had sB-AVR and 141 had TAVR. Valve type did not show a significant association with the risk for myocardial infarction (HR 1.069, 0.837–1.365, $P = 0.59$; Supplementary Material, Fig. 4C, Supplementary Material, Table S13).

Stroke occurred in 1594 patients after the index procedure, 1057 had sB-AVR and 537 had TAVR. Intervention type was not significantly associated with stroke risk (HR 0.908, 0.803–1.027, $P = 0.12$; Supplementary Material, Fig. 4D, Supplementary Material, Table S13).

DISCUSSION

In this retrospective long-term population-based study, undergoing sB-AVR versus TAVR was significantly associated with lower all-cause mortality in patients aged 65 years or older with severe, symptomatic aortic stenosis. For a more accurate comparison of outcomes, we used PSM to compare TAVR and sB-AVR patients who were similar with respect to comorbidities [8] and also identified a survival advantage with sB-AVR. Most other candidate factors, including age, sex, previous heart failure, diabetes and chronic kidney disease, were significantly associated with survival in the overall and PSM cohorts.

These findings are discordant with those of randomized trials in patients with intermediate risk [9] or low risk [10, 11] and with recent meta-analyses [12, 13]. Although randomization allows for control of confounding at baseline, post-randomization exclusion and protocol deviations may still introduce bias [2, 14]. Non-randomized studies using insurance claims databases can be leveraged for real-world evidence, but concerns exist about whether such studies yield unbiased estimates of therapeutic interventions [15]. In RCTs, concomitant procedures (revascularization and others) have not been balanced between patients undergoing TAVR versus sB-AVR, risking performance bias [16, 17]. In the present pragmatic observational study, inclusion of patients with pure symptomatic aortic stenosis without significant coronary artery disease yielded a possibly unique study population for comparing the 2 treatment strategies without the influence of additional procedures, including revascularization. We do not want to conceal that there may be a higher probability of biased results and conclusions when using data derived from accounting information and discharge coding compared with data from prospective clinical databases. However, divergent results between RCTs and real-world studies that address the same scientific question may be disquieting [18], but these divergences demonstrate the need to continue testing assumptions that beneficial treatment effects in RCTs will be borne out in practice [19].

TAVR is associated with lower invasiveness and possibly lower early complication rates and shorter recovery times, suggesting that it might be an option in high-risk and some elderly patients [19]. However, our analyses did not show a survival benefit for elderly patients above 75 years, when selected for TAVR. Remarkably, our separate survival analyses for patients aged 65–75 years and >75 years indicated a significant survival benefit in patients selected for sB-AVR in both groups [4].

We found similar survival with TAVR and sB-AVR up to 1 year, also in accordance with previous randomized studies of high-risk [20, 21] and intermediate-risk patients with 1–5 years of follow-up [9, 22, 23]. Beyond 1 year, however, the Kaplan–Meier curves diverged in favour of sB-AVR, including in our PSM patient comparison. Longer post-procedural observation in previous trials might have revealed similar changes depending on patient selection for either AVR strategy.

As with any new technique or technology, practitioners become more efficient with experience, and outcomes might improve [24]. We performed separate analyses of all-cause mortality in patients treated before and during/after 2015 to take potential improvements and technological innovation into account. Results were similar for both observation periods.

Our findings for outcomes with TAVR versus sB-AVR in 3 subsets of patients with underlying conditions (diabetes, pulmonary diseases, kidney diseases) were consistent with the results for the total study population. Thus, even in patients with these increased risks, long-term outcomes were better in those selected for sB-AVR. In addition, selection for TAVR compared with sB-AVR has consistently been associated with a higher risk of permanent pacemaker implantation. We confirmed this finding, and the higher rate of implantations might have contributed to a higher all-cause mortality rate in patients selected for TAVR [25].

TAVR could be associated with a higher risk of structural degeneration of the prosthetic valve compared with SAVR [26], risking reoperation and worse outcomes for these patients.

We found no significantly higher risk of reoperation with either AVR procedure, however. TAVR is also associated with a specific humoral immune response against alpha-Gal and nonspecific humoral inflammation [27, 28]. A chronic inflammatory response may be associated with increased risk for several conditions that might affect prognosis and overall outcome [29], but the systemic immune response is reported not to differ after TAVR versus sB-AVR [30]. Any contribution of a differential immunologic response to differential outcomes thus is unlikely.

The main strength of this study is the use of a large, representative real-world national database and allocation of patients to AVR procedures essentially based on contemporary international guidelines. The observational design, however, is a limitation, which we sought to mitigate by including several potential confounding factors in the statistical models and performing detailed PSM to support meaningful comparisons. As in any observational research, even with the large sample size and a long-term follow-up in the current work, unmeasured confounding is still possible. Nevertheless, properly adjusted analyses from large registries can support therapeutic decision-making, and our findings undoubtedly reflect real-world results in patients selected for TAVR or sB-AVR.

Limitations

Another limitation is that our data did not include all parameters needed for accurate calculation of surgical risk scores, and we could not calculate frailty scores from our database. Despite obtaining data on further diagnoses and medication, we could not eliminate the possibility that sB-AVR was performed more often in healthier patients. Several subgroup analyses did reveal similar effects in differently characterized cohorts.

Our data do not include a proportion of the most recent prosthetic valve devices used for sB-AVR and TAVR, although we do not expect different results with the latest devices. Furthermore, the data were derived from accounting information and discharge coding, requiring an assumption of correct nationwide coding that cannot be retrospectively verified or corrected. Thus, there is the possibility of biased results and conclusions compared with data from prospective clinical databases.

Although our study is based on a large and representative real-world national database, the study population is drawn exclusively from the Austrian Health Insurance Funds, raising potential concerns about the generalizability and external validity of the findings to a broader patient population. The absence of clinical characteristics may limit the applicability of the results to other healthcare settings and populations and may diminish the broader impact of the study.

Finally, despite coding of aortic stenosis was mandatory to be included in this study, we cannot definitely rule out that some patients had trace or mild accompanying aortic regurgitation in both groups, that was neither the leading diagnosis nor the indication for the intervention.

Conclusions

In conclusion, these mid- to long-term population-based findings suggest that selection for sB-AVR compared with TAVR was significantly associated with lower all-cause mortality in patients aged 65 years or older with severe, symptomatic aortic stenosis. We did not identify a subgroup of patients who had better outcomes with TAVR, but older patients with high surgical risk may still benefit, at least short term, from a less-invasive approach [31, 32]. With its longer-lasting and more durable resolution compared with TAVR, however, sB-AVR may remain the procedure of choice for severe symptomatic aortic stenosis in patients who are surgical candidates. In this group, sB-AVR was associated with significantly lower mortality rates compared with TAVR. Our findings may challenge the current trend of increasing utilization of TAVR in patients who would be appropriate candidates for sB-AVR.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *EJCTS* online.

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Conflict of interest: none declared.

DATA AVAILABILITY

Pseudonymized participant data will be made available upon qualified request starting with publication. Approval of a proposal and a signed data access agreement are mandatory to make data available.

Author contributions

Johann Auer: Conceptualization; Formal analysis; Methodology; Visualization; Writing—original draft; Writing—review and editing. **Pavla Krotka:** Data curation; Formal analysis; Methodology; Software; Validation; Visualization; Writing—original draft; Writing—review and editing. **Berthold Reichardt:** Data curation; Formal analysis; Software; Writing—original draft; Writing—review and editing. **Denise Traxler:** Methodology; Project administration; Validation; Writing—original draft; Writing—review and editing. **Ralph Wendt:** Conceptualization; Methodology; Validation; Writing—original draft; Writing—review and editing. **Michael Mildner:** Software; Visualization; Writing—original draft; Writing—review and editing. **Hendrik Jan Ankersmit:** Conceptualization; Funding acquisition; Methodology; Supervision; Writing—original draft; Writing—review and editing. **Alexandra Graf:** Conceptualization; Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing—original draft; Writing—review and editing.

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